

Goal: Fitting Functions

Given Data

Get Formula

e.g.

Data is periodic

think is periodic

Number born on each day

365 days in a year

Period is 1 year

$V[i][j]$ = # born on day j
in year i

Want Formula

$a \sin t + b \cos t + c$

$t = 2\pi (j/365)$

Don't know a, b, c

Find a, b, c

Use least Squares.

Problem: What are the inputs?

v is an array $N \times 365$

$N = \#$ of years in data set

Least Squares.

$$A x = b$$

$A = \text{matrix}$

$b = \text{vector}$

$x = \text{vector}$

$x = \text{Solutions}$
 $= \text{Parameters}$

$$x = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

`lstsq (A, b)`

output is x

least-squares

Usually $Ax \neq b$

$Ax - b = \text{error}$

Sol. minimizes error

if $\exists x$ s.t. $Ax = b$
then minimizes error

$Ax = b$ No solutions

least squares closest x

$Ax - b$ is small
(hopefully)

e.g. linear regression



$Ax = b$ Unique sol

then $Ax - b = 0$ smallest
poss error

least squares sol

Useful when A is not square

$Ax = b$ ∞ sol's.

least squares gives smallest x